

Dataset Annotation Process Using Roboflow

1. Introduction

This document describes the annotation workflow used to generate a structured object detection dataset for training YOLOv26 models. The goal of annotation was to accurately label traffic-related objects to enable supervised learning for multi-class detection.

2. Dataset Source

The dataset consists of traffic scene images collected from open-source repositories and manually curated frames. Images include varying lighting conditions, object scales, occlusions, and traffic densities to simulate real-world scenarios.

3. Annotation Tool Selection

Roboflow was selected as the annotation platform due to:

- Web-based collaborative annotation
 - Support for YOLO-format bounding boxes
 - Dataset versioning and augmentation tools
 - Automatic export to Ultralytics YOLO format
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4. Annotation Classes

The following object classes were defined:

- SUV
- Car
- Bicycle
- Motorcycle
- Human

Each object instance in an image was annotated with a tight bounding box enclosing the visible object area.

5. Annotation Procedure

1. Images were uploaded to Roboflow as an Object Detection project.
 2. Each image was reviewed individually.
 3. Bounding boxes were manually drawn around all visible objects.
 4. Class labels were assigned per bounding box.
 5. Images containing multiple objects were fully annotated (all visible instances).
 6. Annotation consistency was maintained by following fixed box-tightness rules.
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6. Dataset Splitting

Roboflow automatically split the dataset into:

- Training set ($\approx 70\%$)
- Validation set ($\approx 30\%$)

This ensured that evaluation metrics reflect generalization rather than memorization.

7. Export Format

The annotated dataset was exported in **YOLO format**, producing:

- `/images/train, /images/val`
 - `/labels/train, /labels/val`
 - `data.yaml` defining class names and paths
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8. Annotation Challenges

- Small objects (bicycles, motorcycles)
- Partial occlusions
- Overlapping objects
- Background clutter

These challenges motivated later quality assurance and model analysis.

9. Conclusion

The annotation process produced a structured and YOLOv26-compatible dataset suitable for training and evaluating object detection models under realistic traffic conditions.

